Meta Optimal Transport

Learn to solve optimal transport problems

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Contents

Short optimal transport introduction

What is optimal transport?

- Optimal transport is used to **compare probability distributions**

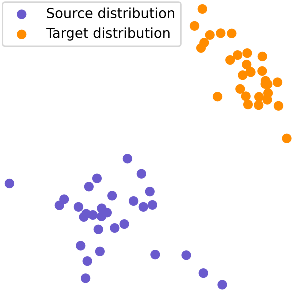


Figure: Discrete distribution

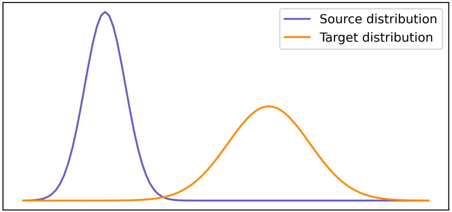
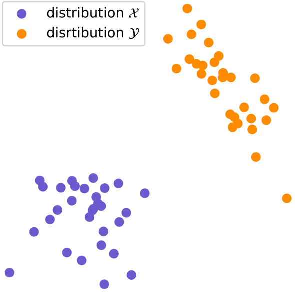
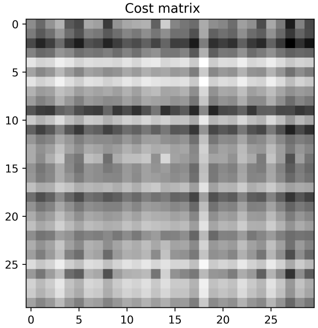


Figure: Continuous distribution

Discrete probability distributions

- Two sets of data points $x_1, ..., x_n \in \mathbb{R}^d$ and $y_1, ..., y_m \in \mathbb{R}^d$
- Probability distributions $\mathcal{X} = \sum_{i=1}^n \alpha_i \delta_{x_i}$ and $\mathcal{Y} = \sum_{i=1}^m \beta_i \delta_{y_i}$
- Cost matrix $C_{i,j} = d(x_i, y_j)$ for some distance d.

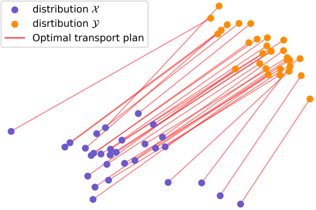


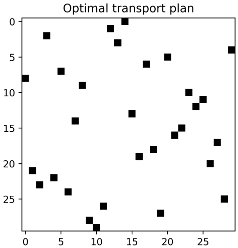


Kantorovich problem

- The Kantorovich optimal transport problem looks for an optimal transport plans $P \in \Pi_{\alpha,\beta}$, for $\Pi_{\alpha,\beta} = \{P \in \mathbb{R}_+^{n\times m} | P\mathbb{1}_m = \alpha, P^T\mathbb{1}_n = \beta\}$ such that

$$P = \arg\min_{P\in\Pi_{\alpha,\beta}} \sum_{\substack{i=1,\dots,n\\ j=1,\dots,m}} P_{i,j}C_{i,j}$$



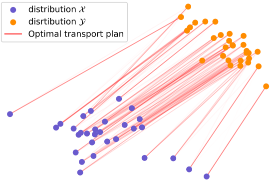


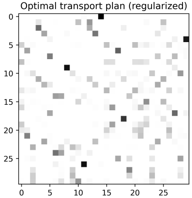
Regularized Kantorovich problem [Cuturi 2013]

- Add an entropy regularization term: $H(P) = -\sum_{i,j} P_{i,j}(\log P_{i,j} - 1)$

$$P = \arg\min_{P \in \Pi_{\alpha,\beta}} \sum_{\substack{i=1,\ldots,n \\ j=1,\ldots,m}} P_{i,j} C_{i,j} - \epsilon\, H(P)\ ,$$

- **Why regularize ?** Strictly convex optimization problem, fast algorithms.
- Loses sparsity on the optimal transport plan.





Solving the entropy regularized OT problem

- **Primal problem:**

$$P^* \in \arg\min_{P \in \Pi_{\alpha,\beta}} \langle C, P \rangle - \epsilon H(P)$$

- **Lagrangian:**

$$\mathcal{L}(P, f, g) = \sum_{i,j} P_{ij} C_{ij} + \epsilon P_{ij} \big(\log P_{ij} - 1\big) + f^\top (P \mathbf{1}_n - \alpha) + g^\top (P^\top \mathbf{1}_m - \beta)$$

$$\frac{\partial \mathcal{L}(P, f, g)}{\partial P_{ij}} = C_{ij} + \epsilon \log P_{ij} + f_i + g_j$$

$$\frac{\partial \mathcal{L}(P, f, g)}{\partial P_{ij}} = 0 \quad \Longrightarrow \quad P_{ij} = \exp\left(\frac{f_i}{\epsilon}\right) \exp\left(-\frac{C_{ij}}{\epsilon}\right) \exp\left(\frac{g_j}{\epsilon}\right)$$

Solving the entropy regularized OT problem

- The solution of the entropy regularized transport is of the form

$$P^* = \mathrm{diag}(u) K \mathrm{diag}(v),$$

  for $K_{i,j} = e^{-C_{i,j}/\epsilon}$
- Relationship with the dual variables:

$$u_i = e^{f_i/\epsilon}, \quad v_j = e^{g_j/\epsilon}$$

- **Sinkhorn algorithm** $\mathrm{diag}(u)$ and $\mathrm{diag}(v)$ exist and are unique

Sinkhorn algorithm

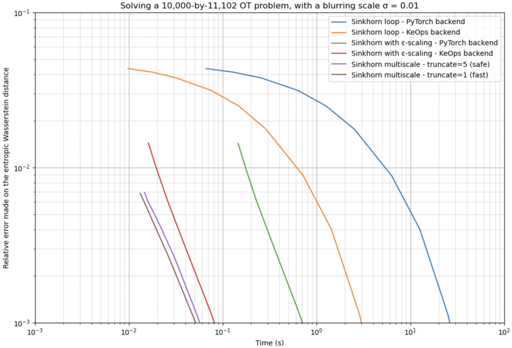
Input: Cost matrix C, marginals α, β, regularization ε, iterations T
Output: Transport plan P
Compute $K \leftarrow \exp(-C/\varepsilon)$
Initialize $u_0 \leftarrow 1_n$
while $t \leq T - 1$ **do**
  $v_i \leftarrow \beta \oslash K^T u_{i-1}$ for $\oslash$ the elementwise division
  $u_i \leftarrow \alpha \oslash K v_i$
  $t \leftarrow t + 1$;
end
Compute $P \leftarrow \text{diag}(u^T) K \text{diag}(v^T)$

- The algorithm alternatively scales the rows and the columns to match the marginals:

$$P \mathbb{1}_m = \alpha, P^T \mathbb{1}_n = \beta$$

Sinkhorn algorithm

- Complexity: $O(Tnm)$.
- GPU friendly: all rows and columns can be updated in parallel
- Memory complexity: need to store K of size $n \times m$. Can be lazily computed with geomloss library from Feydy et al. 2019.



Continuous optimal transport

- Given two continuous measures α, β in $\mathcal{X} = \mathcal{Y} = \mathbb{R}^d$, the square Wasserstein distance is

$$W_2^2(\alpha,\beta) := \min_{\pi \in \Pi(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|_2^2 \, d\pi(x,y) \tag{1}$$

$$= \min_{T} \int_{\mathcal{X}} \|x-T(x)\|_2^2 \, d\alpha(x) \tag{2}$$

where T is a transport map pushing α to β, i.e. $T_{\#}\alpha = \beta$ where $T_{\#}\alpha(B) := \alpha(T^{-1}(B))$ and $\Pi(\alpha,\beta) = \{\pi \in \mathcal{M}(\mathcal{X}\times\mathcal{Y}), P_{\mathcal{X}\#}\pi = \alpha, P_{\mathcal{Y}\#}\pi = \beta\}$

Convex dual potentials

- **Dual formulation** We look for the optimal potential ψ

$$\psi^*(.;\alpha,\beta) \in \arg\max_{\psi \text{ convex}} \int_{\mathcal{X}} \psi(x) d\alpha(x) + \int_{\mathcal{Y}} \overline{\psi}(y) d\beta(y)$$

where $\overline{\psi}$ is the convex conjugate of ψ defined as

$$\overline{\psi}(y) := \sup_{x \in \mathcal{X}} \langle x, y \rangle - \psi(x)$$

- **Recovering the primal solution (Brenier, 1991)** The optimal transport map is obtained as

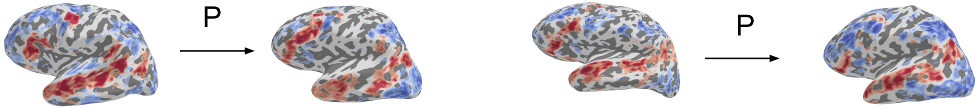
$$T^*(x) = \nabla_x \psi^*(x)$$

Efficiently estimating OT maps

- Neural OT models Korotin et al. 2019: ICNNs to estimate maps between continuous high-dimensional measures given samples from these.
- Scetbon et al. 2021 leverage structural assumptions on coupling and cost matrices to reduce complexity.
- Courty et al. 2017 learn a latent space in which the Euclidian distance between the measure's embeddings is equivalent to the Wasserstein distance.
- Lacombe et al. 2023 learn to generate wasserstein barycenters of images.

MetaOT: Solving many OT problems efficiently

- In many applications, we need to solve multiple OT problems
- **Example**: brain alignement over a vast group of subjects



- For one given OT problem, existing methods are efficient (e.g. Sinkhorn)
- **MetaOT**: Can we leverage the joint structure of a dataset to speed up the computation of OT plans across the dataset?

Amortized optimization

Optimization problems are often repeatedly solved

- Many problems require solving similar optimization problems repeatedly over different contexts.

$$y^*(x) \in \arg\min f(y;\ x\ )$$

context

- **Example**: Reinforcement learning

state

$$a^*(s) \in \arg\max Q(\ s\ ;\ a\ )$$

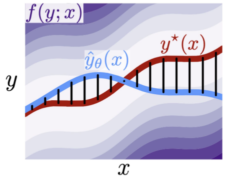
action

- Optimization can be computationally expensive and re-optimizing from scratch each time is sub optimal.
- **Idea** Amos et al. 2023: learn to uncover the shared structure between optimization problems by learning $\hat{y}_\theta(x) \approx y^*(x)$

Learning the amortized model

Regression based

$$\mathcal{L}(\hat{y}_\theta) := \mathbb{E}_{x \sim p_X}\left[\hat{y}_\theta(x) - y^*(x)\right]_2^2$$

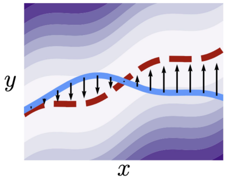


→ requires optimal solutions for training, can be computationally expensive

Figures from Amos, 2022

Objective based

$$\mathcal{L}(\hat{y}_\theta) := \mathbb{E}_{x \sim p_X}\left[f(\hat{y}_\theta(x); x)\right]$$

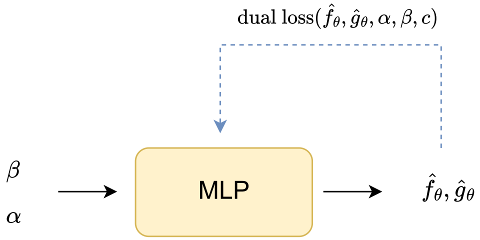


→ does not require optimal solutions for training!

Meta optimal transport between discrete measures

Discrete MetaOT overview

- Fix cost c and sample discrete measures from a distribution $(\alpha_i, \beta_i) \sim \mathcal{D}$ of OT problems.
- Predict dual potentials with a model $\hat{f}_\theta, \hat{g}_\theta$.
- Backpropagation by minimizing the dual loss.



Meta optimal transport objective

- Optimal dual potentials

$$f^*, g^* \in \arg \max_{f \in \mathbb{R}^n, g \in \mathbb{R}^m} \langle a, f \rangle + \langle b, g \rangle - \epsilon \langle \exp(f/\epsilon), K \exp(g/\epsilon) \rangle$$

- **Mapping between the duals** First order condition provides an equivalence between f and g at optimality

$$g(f; b, c) := \epsilon \log b - \epsilon \log \left(K^T \exp\{f/\epsilon\} \right).$$

  Hence, it is sufficient to predict one of the potentials, e.g. f^*.

- **MetaOT objective**:

$$f^*(\alpha, \beta, c, \epsilon) \in \arg \min_{f \in \mathbb{R}^n} J(f; \alpha, \beta, c)$$

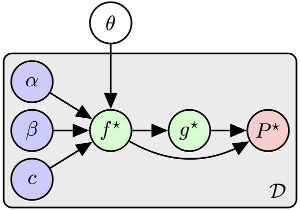
$$\text{for } -J(f; \alpha, \beta, c) := \langle \alpha, f \rangle + \langle \beta, g \rangle - \epsilon \langle \exp(f/\epsilon), K \exp(g/\epsilon) \rangle$$

Amortized model

- Predict solution to $f^*(\alpha, \beta, c, \epsilon) \in \arg\min_{f \in \mathbb{R}^n} J(f; \alpha, \beta, c)$ with $\hat{f}_\theta(\alpha, \beta, c)$ parametrized by θ, by mimimizing

$$\mathbf{E}_{\alpha,\beta\sim\mathcal{D}}[J(\hat{f}_\theta(\alpha, \beta, c); \alpha, \beta, c)]$$

- The model $\hat{f}_\theta(\alpha, \beta, c)$ is a fully connected MLP mapping from the atoms of the measures to the duals.
- Recover $\hat{P}^*_\theta = \text{diag}(e^{\hat{f}_\theta/\epsilon}) K \text{diag}(e^{\hat{g}_\theta/\epsilon})$



Discrete (Entropic)

Results: grayscale images transport

- **Grayscale images** A pair of images α_0 and α_1 is cast as a discrete 2d measure where the intensities define the probabilities of the measures.
- **Goal** Optimal transport interpolation between the two measures
- MetaOT is $\simeq 100$ times computationnaly cheaper and produces a nearly identical coupling

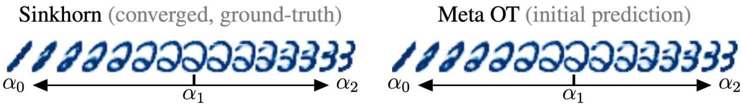


Figure: Interpolation between test set digits

Results: spherical transport problem

- **Spherical transport problem** Transport problem between supply and demand locations. The transport cost is given by the spherical distance $c(x, y) = \arccos(\langle x, y \rangle)$
- Meta OT's prediction (right) is $\simeq 37500$ times faster than solving Sinkhorn to optimality and is close to the optimal maps obtained with Sinkhorn (left).

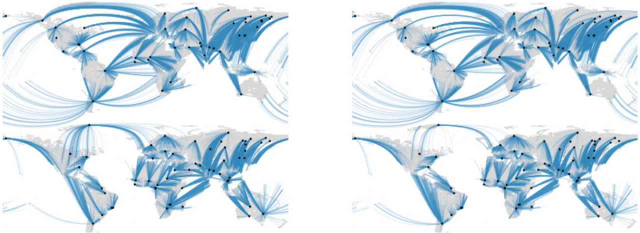


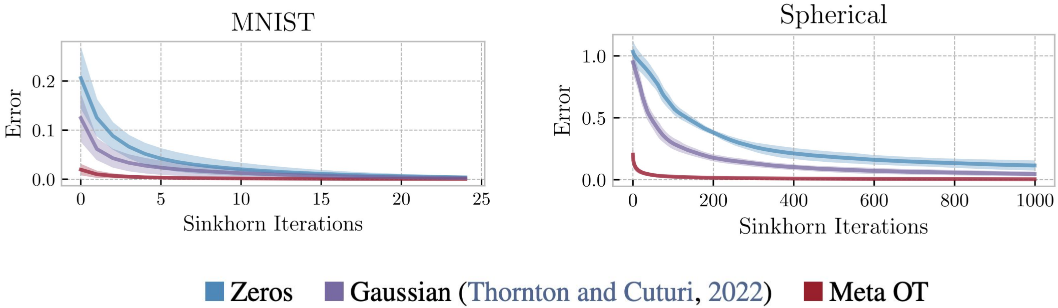
Figure: Test set coupling predictions. Supply locations in black dots and blue lines show the sperical transport maps.

MetaOT as initialization of Sinkhorn

- Sinkhorn algorithm can converge faster if well initialized.
- Typical intializations: $u^0 = \mathbf{1}$, Gaussian $\to$ ignore structure of the problem.
- MetaOT predicts $\hat{f}_\theta(\alpha, \beta, c)$, $\hat{g}_\theta(\alpha, \beta, c)$.
  Initialize Sinkhorn with $u^0 = e^{\hat{f}_\theta/\epsilon}$.

Method	**Cost per OT problem**
Sinkhorn (from scratch)	$O(T\, nm)$
MetaOT only	$O(n + m)$
MetaOT + Sinkhorn	$O(T_{\text{small}}\, nm)$

MetaOT as initialization of Sinkhorn



Out of distribution generalization

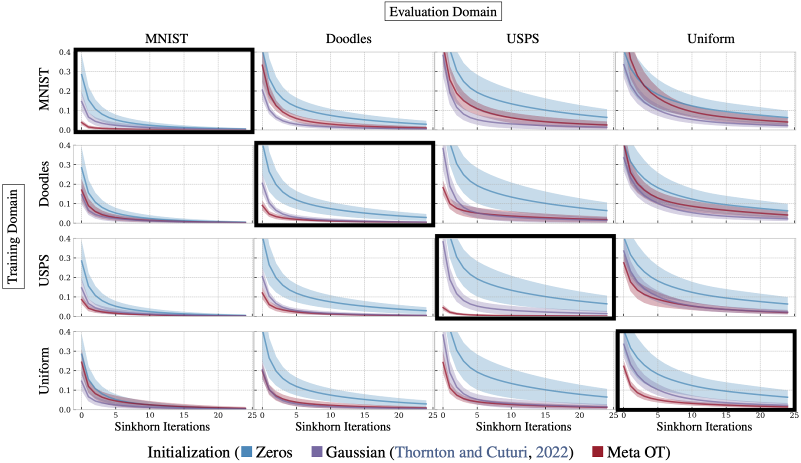
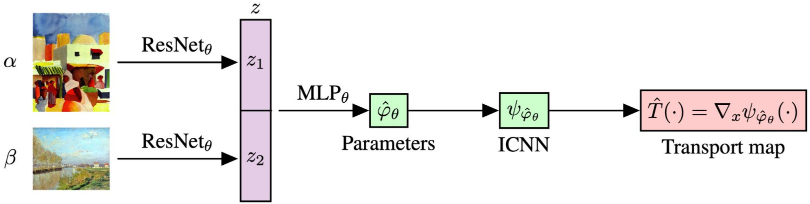


Figure 10. Cross-domain experiments evaluating how well a model trained on one dataset generalizes to another dataset. Notably, we are able to train only on a uniform distribution and transfer reasonable initializations to the image datasets. This indicates that training larger-scale Meta OT models for more general classes of discrete OT problems may be able to provide a fast and reasonable initialization.

Meta optimal transport between continuous measures

Continuous MetaOT overview

- Discrete MetaOT $\rightarrow$ predict dual potentials.
- Continuous MetaOT $\rightarrow$ predict Input Convex Neural Networks (ICNN) parameters:
  - Dual potential is a convex function.
  - ICNNs parametrize convex functions.



Continuous optimal transport

- **Dual formulation** We look for the optimal potential ψ^*

$$\psi^*(.;\alpha,\beta) \in \arg\max_{\psi \text{ convex}} \int_{\mathcal{X}} \psi(x) d\alpha(x) + \int_{\mathcal{Y}} \overline{\psi}(y) d\beta(y)$$

  where $\overline{\psi}$ is the convex conjugate of ψ defined $\overline{\psi}(y) := \sup_{x\in\mathcal{X}} \langle x, y\rangle - \psi(x)$
- **(Brenier, 1991)** Optimal transport map: $T^*(x) = \nabla_x \psi^*(x)$
- **Input convex neural networks (ICNN)**: the output of the neural network is a convex function of (some of) the inputs.

- **Idea:** ICNN learns convex potentials $\to$ gradient gives map

Input convex neural networks (ICNN)

- ICNN: Neural networks $f(.,\theta)$ parametrized by θ such that the function $x \mapsto f(x,\theta)$ is convex.

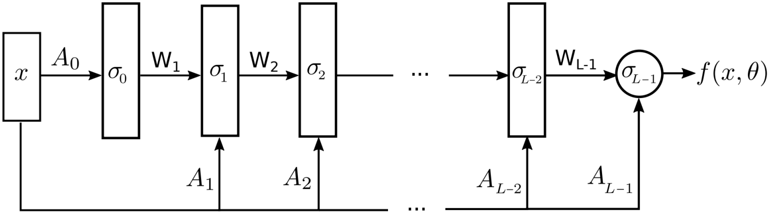


Figure 2. The input convex neural network (ICNN) architecture.

Wasserstein-2 Generative Networks (W2GN) [Korotin et al. 2019]

- Model ψ_φ and $\overline{\psi_\varphi}$ with two separate ICNNs parametrized by φ.
- Wasserstein-2 Generative Networks loss

dual objective

$$\mathcal{L}(\varphi;\alpha,\beta) := \mathbb{E}_{x\sim\alpha}\left[\psi_\varphi(x)\right] + \mathbb{E}_{y\sim\beta}\left[\langle \nabla\overline{\psi_{\cancel{\varphi}}}(y), y\rangle - \psi_\varphi(\nabla\overline{\psi_{\cancel{\varphi}}}(y))\right]$$

$$+\ \gamma\,\mathbb{E}_{y\sim\beta}\left\|\nabla\psi_\varphi \circ \nabla\overline{\psi_\varphi}(y) - y\right\|_2^2$$

cycle consistency regularization

where $\cancel{\varphi}$ is a detached copy of the parameters.

Wasserstein-2 Generative Networks (W2GN) [Korotin et al. 2019]

Input: α, β, φ_0
Output: Transport map T
while $t \leq N - 1$ **do**
  Sample from (α, β) and estimate $\mathcal{L}(\varphi_{i-1})$
  Update φ_i with approximation to $\nabla_\phi \mathcal{L}(\varphi_{i-1})$
end
Compute $T \leftarrow \nabla_x \psi_{\varphi_N}(.)$

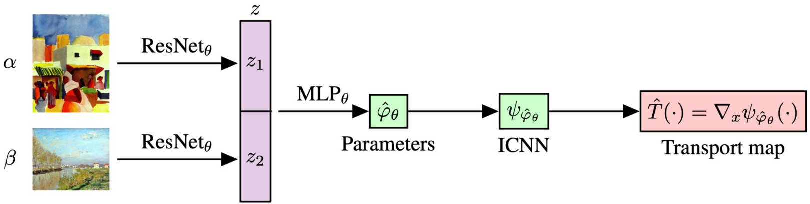
Algorithm 1: W2GN$(\alpha, \beta, \varphi_0)$

- Here, T is optimized for each pair of measures (α, β) from scratch.

Meta ICNN

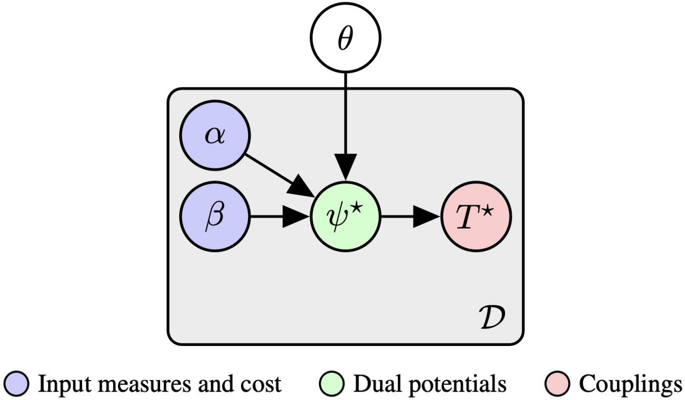
- Meta ICNN predicts the parameters φ of and ICNN ψ_φ that approximates the dual convex potentials by minimizing the W2GN loss.

$$\min_\theta \mathbb{E}_{(\alpha,\beta)\sim\mathcal{D}} \mathcal{L}(\hat{\varphi}_\theta; \alpha, \beta).$$



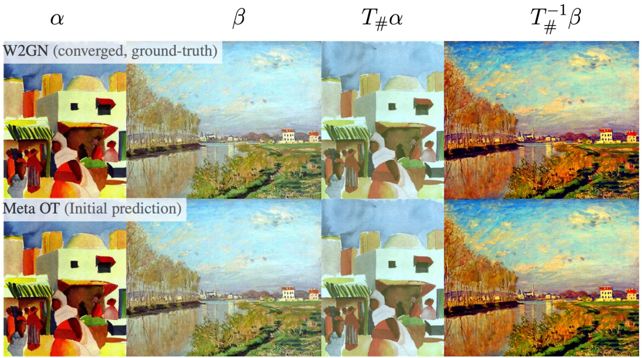
Meta ICNN

- As for discrete MetaOT, MetaICNN does not require ground truth solutions $\varphi^*$for training.



Results on color transfer

- **Color transfer** Color transfer between images can be casted as an optimal transport problem between 3D continuous color distributions.
- MetaOT's prediction is $\simeq 1000$ times faster than training W2GN from scratch.



Meta ICNN as initilisation for W2GN

- **Color transfer** Color transfer between images can be casted as an optimal transport problem between 3D continuous color distributions.
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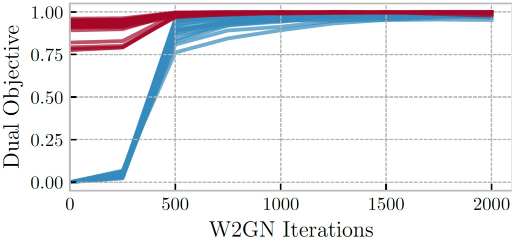
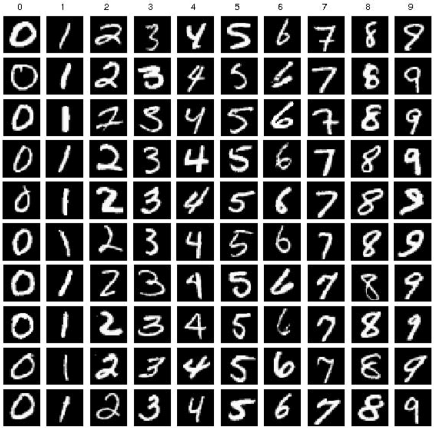


	Iter	Runtime (s)	Dual Value
Meta OT + W2GN	None	$3.5 \cdot 10^{-3} \pm 2.7 \cdot 10^{-4}$	$0.90 \pm 6.08 \cdot 10^{-2}$
	1k	$0.93 \pm 2.27 \cdot 10^{-2}$	$1.0 \pm 2.57 \cdot 10^{-3}$
	2k	$1.84 \pm 3.78 \cdot 10^{-2}$	$1.0 \pm 5.30 \cdot 10^{-3}$
W2GN	1k	$0.90 \pm 1.62 \cdot 10^{-2}$	$0.96 \pm 2.62 \cdot 10^{-2}$
	2k	$1.81 \pm 3.05 \cdot 10^{-2}$	$0.99 \pm 1.14 \cdot 10^{-2}$

Conclusion

- **When MetaOT helps**
  - Many OT problems with a shared structure, similar supports/costs
  - All the OT problems share the same regularization parameter ϵ
- **When MetaOT does not help**
  - Single OT problem
  - Many OT problems with very different supports/costs
  - Large domain shifts



Conclusion

- MetaOT speeds up OT plan calculations by multiple orders of magnitude on datasets with shared structure.
- **Limitations**
  - Meta OT does not make previously intractable problems tractable.
  - Out of distribution generalization is not guaranteed.
  - Need to train MetaOT for each value of regularization parameter ϵ.
- **Extensions**
  - Learn to predict other OT problems: Fused Unbalanced Gromov-Wasserstein [Mazelet et al. 2025], graph node permutation with Sinkhorn [Krzakala et al. 2025].

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